
*Q-sensor bracelet (Credit: www.sporttechie.com)*

municate non-verbally. Sensors in the garment respond to the wearer's pulse, muscle tension and proximity, converting physical information into coloured light animations that emulate different emotions like a pounding heart, rush of adrenaline and the feeling of butterflies in the stomach. The device will help minimise injury and provide parents with a convenient, calming and socially-acceptable way of tracking and managing their child's behaviour.

- A device created at Massachusetts Institute of Technology (MIT) emits radio signals to reflect off a person's front and back. By measuring heartbeat and breathing, the device can accurately detect the person's emotional state. Such remote emotion-sensing technology could be used to diagnose or track conditions such as depression and anxiety, as well as for non-invasive health monitoring and diagnosis of heart conditions. The device, named EQ-Radio, has been found to be 87 per cent accurate at detecting whether a person is excited, happy, angry or sad—all without on-body sensors or facial-recognition software.
- Auto-emotive is a project of MIT's Affective Computing Group, which is based on the exploration of the potential of emotional connection to machines with humans. The goal of the project is to implement mood recognition technology in cars, to create cars that can empathise with the driver and improve the experience of being behind the wheel.
- Development of emotional AI will lead to more effective robotic personal assistants. IBM is developing techniques to add human-like qualities to robotic systems. Such robots will be able to distinguish between, and react to, different people and their emotional states.
- A new sensing technology developed by Panasonic Corp. can detect a person's emotions in a contactless manner. This information can be used for predicting a driver's drowsiness and help keep him or her awake.
- Some of the world's elite coaches, teams and individual athletes have used headsets produced by SenseLabs Inc., USA, which can be connected to an iPhone or iPad via Bluetooth and have dry sensors for assessing brain performance. This sensing technology makes it possible to identify the strengths and weaknesses in problem-solving, multitasking, resource management, decision-making and sleep tendencies. The headsets also provide customised exercise protocols to users to improve their mental facuity, concentration and sleep.
- Bracelets that can measure emotions via electro-dermal activity have been developed and commercialised by many companies. In these devices, the display reflects the user's heightened emotional states. Q-sensor developed by Affectiva, USA, detects and records physiological signs of stress and excitement by measuring slight electrical changes in conductance and temperature of skin.

## Expected benefits

While applications and benefits of emotion-sensing technology are limited only by our imagination, there are some obvious benefits.

- Emotion-sensing technologies can help employees make better decisions, improve their focus and per-